Mehdi Mohebbi

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Research Interests: Deep learning architectures, Representation learning and latent spaces, Continual learning, Sequence and time-series foundation models, Machine learning for health and wearable sensing

Education

University of Tehran Sep 2024 – expected Jun 2027

M.Sc. in Computer Science (Artificial Intelligence and Soft Computing)

Thesis: *Document-Level Offline Handwritten Text Recognition using Deep Learning Approaches*

Advisor: Dr. Bagher BabaAli | GPA: 17.83/20 (3.69/4)

Kharazmi University Sep 2020 – Jul 2024

B.Sc. in Computer Science

GPA: 19.34/20 (3.98/4)

Honors & Awards

1st Place, Sussex-Huawei Locomotion (SHL) Challenge 2026 Oct 2026

HASCA Workshop at ACM UbiComp/ISWC 2026

1st Place, Sussex-Huawei Locomotion (SHL) Challenge 2025 (Task 1) Oct 2025

HASCA Workshop at ACM UbiComp/ISWC 2025

1st Place, 4th National Student Operations Research Competition Jul 2024

Held at the 17th International Conference of the Iranian Operations Research Society

Research Experience

Research Assistant Jun 2025 – Present

University of Tehran

Supervisor: Dr. Bagher BabaAli

- Designing compact handwritten text recognition models for English and Persian using JEPA-style latent prediction.
- Working with wearable and smartphone motion-sensor (IMU) data for human activity recognition and neurological disorder diagnosis.
- Built the 1st-place SHL Challenge entries for smartphone-sensor locomotion recognition: in 2025, an ensemble of pre-trained time-series foundation models and biosignal models trained from scratch; in 2026, early and late fusion of frozen time-series foundation models.

Research Assistant May 2023 – Present

Kharazmi University

Supervisor: Dr. Bardia Panahbehagh

- Research on Graphical Finite Population Sampling (GFS); co-developed an open-source Python package that implements and extends it (github.com/mehdimhb/graphical-sampling).
- Introduced Intelligent n-Means Spatial Sampling, a GFS-based method for designing well-spread samples that give more precise estimates in spatial populations.
- Proposed a new index for measuring the spatial spread of samples.
- Developing deep learning models to learn and optimize the space of sampling designs for given populations.

Publications

Journal Articles

[J1] **Mehdi Mohebbi**, Amir Mohammad Basuti, Bahar Marashi, Hasti Jahantigh, Mohammad Mahdi Azizi, Melika Khosravi, Bagher BabaAli. "FallPGR: Proposal grouping and refinement for temporal fall localization in continuous multi-radar UWB streams." *International Journal of Activity and Behavior Computing (IJABC)*, 2027. [Under review] Challenge paper for the ABC 2027 UWB FallTrack Challenge.

[J2] Bardia Panahbehagh, **Mehdi Mohebbi**, Amir Mohammad HosseiniNasab. "Measuring spatial spread via n-means balanced clustering." *Journal of Survey Statistics and Methodology (JSSAM)*, 2026. [In press] An earlier version, combined with "Intelligent n-means spatial sampling", is available as arXiv:2510.24183 (2025). [link]

[J3] Bardia Panahbehagh, **Mehdi Mohebbi**. "Intelligent n-means spatial sampling." *Journal of Survey Statistics and Methodology (JSSAM)*, 2026. [Under revision] Minor revision, third review round. An earlier version, combined with "Measuring spatial spread via n-means balanced clustering", is available as arXiv:2510.24183 (2025). [link]

Conference and Workshop Papers

[C1] **Mehdi Mohebbi**, Amir Basuti, Mohammad Mahdi Azizi, Bagher BabaAli. "Leveraging frozen time-series foundation models for IMU-based freezing of gait detection." *33rd National and 11th International Iranian Conference on Biomedical Engineering (ICBME)*, 2026. [Accepted] To appear in IEEE Xplore.

[C2] **Mehdi Mohebbi**, Ali Majlesi, Amir Basuti, Iraj Gharaee, Mohammad Mahdi Azizi, Bagher BabaAli, Pouya Mohseni. "Early and late fusion of frozen foundation models for IMU-based human activity recognition." *Sussex-Huawei Locomotion (SHL) Challenge 2026, Companion of the ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC)*, 2026. [Accepted] Equal contribution (first two authors).

[C3] Ali Majlesi, Amir Basuti, **Mehdi Mohebbi**, Iraj Gharaee, Bagher BabaAli. "Phase-aware frozen time-series foundation models for parameter-efficient clinical gait assessment." *IEEE Signal Processing in Medicine and Biology Symposium (SPMB)*, 2026. [Accepted]

[C4] Vincent Loonis, Bardia Panahbehagh, Amir E. Ghara, **Mehdi Mohebbi**. "Artificial bee colony improvement of determinantal sampling designs." *18th Iranian Statistics Conference, K.N. Toosi University of Technology, Tehran, Iran*, 2026.

[C5] Mohammad Foad Abdi, Yousef Alikhani, Mohammad Mahdi Azizi, Bagher BabaAli, **Mehdi Mohebbi**, Arash Nasr Esfahani. "Ensemble of foundation models for sensor-based locomotion and transportation mode recognition." *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC '25), pp. 949-955*, 2025. [link]

[C6] Bardia Panahbehagh, **Mehdi Mohebbi**, Amir Mohammad HosseiniNasab. "A density-based index to measure spread of a spatial sample." *Proceedings of the 2nd International Conference on Mathematics and its Applications, Shahid Chamran University of Ahvaz, Ahvaz, Iran, pp. 6-13*, 2025.

[C7] Bardia Panahbehagh, Amirhosein Javadpour, Zohre Bagherpour, Hossein Kian, **Mehdi Mohebbi**. "Design strategies for training samples in regression." *Proceedings of the 2nd International Conference on Mathematics and its Applications, Shahid Chamran University of Ahvaz, Ahvaz, Iran, pp. 14-22*, 2025.

[C8] Mahdi Cherakhloo, Mahtab Ranjbar, **Mehdi Mohebbi**, Bijan Vosoughi Vahdat. "DCT-HistoTransformer: Efficient lightweight vision transformer with DCT integration for histopathological image analysis." *31st National and 9th International Iranian Conference on Biomedical Engineering (ICBME), pp. 409-415*, 2024. [link]

Teaching Experience

Teaching Assistant Sep 2026 – Present

Kharazmi University

- Probability and Statistics for master's international students (Fall 2026), Dr. Bardia Panahbehagh.

Teaching Assistant Sep 2025 – Jun 2026

University of Tehran

- Machine Learning (Fall 2025), Dr. Bagher BabaAli.
- Natural Language Processing (Spring 2026), Dr. Bagher BabaAli.

Teaching Assistant Feb 2023 – Jun 2024

Kharazmi University

- Principles of Theory of Computation (Spring 2023), Dr. Arash Ahadi.
- Artificial Intelligence (Fall 2023), Dr. Mahdi Cherakhloo.
- Foundations of Numerical Analysis (Fall 2023), Dr. Jamshid Saeidian.
- Stochastic Processes (Spring 2024), Dr. Bardia Panahbehagh.

Professional Experience

Statistical Consultant (part-time) Jan 2025 – Jun 2025

Divar

- Designed and conducted a market research study for Divar with Dr. Bardia Panahbehagh, including the sampling plan and survey methodology for collecting representative market data.

Projects

graphical-sampling — Open-source Python package that implements and extends Graphical Finite Population Sampling, including Intelligent n-Means Spatial Sampling for designing well-spread samples and a Density Index for measuring the spatial spread of a sample. 2023

shl2026 — Code for "Early and late fusion of frozen foundation models for IMU-based human activity recognition" (SHL Challenge 2026, 1st place): combining frozen time-series foundation models for locomotion and transportation mode recognition from smartphone motion sensors. 2026

fog-tsfm — Code for "Leveraging frozen time-series foundation models for IMU-based freezing of gait detection" (ICBME 2026): detecting freezing of gait in Parkinson's disease from wearable IMU data with frozen foundation models. 2026

Skills

Programming: Python, SQL

ML / Scientific: PyTorch, Lightning, NumPy, scikit-learn

Tools: Git, Linux, LaTeX

Languages

Languages: Persian (Native), English (Fluent, C1)